

A05 Reflection Journal

Analyzing *Arrival* Through the Lens of NLP

Team: DataProxy

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As Team DataProxy, Jonah Joseph, Fifehanmi Ogedengbe, Justin Davis, Brandon Matias, Samantha Mireles, and Elijah Raines, we learned that understanding language takes more than translating one word at a time. In *Arrival*, the word "weapon" causes fear because the humans do not know whether the heptapods mean something dangerous or a tool meant to help them. This example showed us how context, culture, and intention can change the meaning of a message. We learned that good communication requires patience and checking what someone means before making conclusions (Villeneuve, 2016).

We also learned from Louise's process that language has to be built step by step. She starts with simple words, points to herself, uses visual symbols, repeats ideas, and looks for patterns in the circular writing. Connecting those scenes to NLP helped us understand the differences between rule-based, statistical, and deep-learning approaches. Rule-based NLP follows written rules, statistical NLP compares patterns across examples, and deep NLP learns more complex relationships from large amounts of data. Even though the movie does not name these technologies, the comparison helped us see why a system needs enough examples and context before it can give a reliable interpretation (Stryker & Holdsworth, 2024; Villeneuve, 2016).

Overall, this project taught Team DataProxy that machine translation, pattern recognition, and language models can support communication, but they cannot replace human judgment. A system may recognize symbols or suggest a translation while missing emotion, cultural differences, or the speaker's real purpose. The confusion in *Arrival* showed how one uncertain interpretation can affect everyone when people react too quickly. Our main takeaway is that NLP works best when people question unclear results, share information, and use technology as a tool instead of assuming its first answer is always correct (Stryker & Holdsworth, 2024; Villeneuve, 2016).

References

- Stryker, C., & Holdsworth, J. (2024, August 11). *What is NLP (natural language processing)?* IBM.
<https://www.ibm.com/think/topics/natural-language-processing>
- Villeneuve, D. (Director). (2016). *Arrival* [Film]. Paramount Pictures.